Paleomagnetism of Quaternary and late Tertiary sediments on Mokowan Butte, southwestern Alberta

R. W. BARENDREGT
Department of Geography, The University of Lethbridge, Lethbridge, Alta., Canada T1K 3M4
E. IRVING
Geological Survey of Canada, Pacific Geoscience Centre, Sidney, B.C., Canada V8L 4B2
AND
E. T. KARLSTROM
Department of Geography, California State University Stanislaus, Turlock, CA 95380, U.S.A.

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Mokowan Butte in southwestern Alberta is capped by 32 m of unconsolidated sediment comprising five tills and their associated paleosols. Previous stratigraphic work has indicated that the older tills are probably Early Pleistocene to late Tertiary in age. Paleomagnetic studies, based on 125 samples, show that the uppermost till–paleosol unit and the modern subsoil are normally magnetized, whereas the till–paleosol units from the middle of the sequence have reversed magnetization. We argue that these belong to the Brunhes and Matuyama polarity zones, respectively. The lowest part of the sequence has not been studied in detail and is generally too coarse for paleomagnetic sampling, and samples that have been obtained contain no accurate record of the paleofield. These new data, together with those from Wellsch Valley in Saskatchewan, Banks Island in the Northwest Territories, Fort Selkirk in the Yukon, and Merritt in British Columbia, bring to five the number of early Quaternary sequences in Canada that have yielded a record of glacial sediments deposited during the Matuyama chron, and hence record glacial events prior to the last reversal of the geomagnetic field (720 ka).

La butte de Mokowan, dans le sud-ouest de l'Alberta, est coiffée par 32 m de sédiments meubles incluant cinq dépôts morainiques et leurs paléosols associés. Selon les études stratigraphiques antérieures, les plus anciens dépôts morainiques seraient d'âge pléistocène précoce à tertiaire tardif. Les mesures paléomagnétiques, sur 125 échantillons, révèlent que l'unité sommitale moraine–paléosol et le sous-sol actuel sont caractérisés par une aimantation normale, tandis que les unités moraine–paléosol du milieu de la séquence montrent une aimantation inversée. Nous croyons que ces unités correspondent aux magnétozones de Brunhes et de Matuyama, respectivement. La partie inférieure de la séquence n'a pas été étudiée en détail, elle est en général formée de matériaux trop grossiers pour un échantillonnage paléomagnétique adéquat, et les échantillons prélevés n'ont pu fournir des résultats précis sur le paléochamp. Ces nouvelles données, ajoutées à celles connues de la vallée Wellsch en Saskatchewan, de l'île Banks du Territoire du Nord-Ouest, de Fort Selkirk du Yukon et de Merritt de la Colombie-Britannique, portent à cinq le nombre de séquences du Quaternaire précoce au Canada qui fournissent un registre de dépôts de sédiments glaciaires datant de l'époque de Matuyama, et par conséquent elles documentent des évènements glaciaires antérieurs à la dernière inversion du champ magnétique terrestre (720 ka).

[Traduit par la rédaction]

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Introduction

Stratigraphic work on unconsolidated sediments that cap Mokowan Butte in southwestern Alberta has identified five till units on each of which paleosols were developed (Karlstrom 1987, 1988*a*, 1988*b*). These sediments are the northernmost remnant (Fig. 1) of the "Kennedy drift" of Alden and Stebinger (1913) and Horberg (1956). Mokowan Butte was chosen for this study because it provided the most complete stratigraphy, it was the most accessible, and the materials were generally finer grained than other exposures of Kennedy drift. The area is outside the limits of Wisconsin glaciation, thus the tills are all pre-Wisconsin in age (older than 65 ka). Using various soil properties, Karlstrom (1987, 1988*a*, 1988*b*, 1991) concluded that the paleosols formed under interglacial conditions that were warmer and wetter than the present. Hence at least five glacial and five interglacial intervals are recorded in the Mokowan Butte sediments. Karlstrom numbered the till units and associated paleosols, recording these events 1–5, number 1 being the oldest. He also estimated the time taken for each paleosol to develop, and by correlation with dated sequences in the midcontinental U.S.A. (Boelstorff 1978) proposed that tills 1 and 2 and their associated paleosols and till 3 were older than 2.2 Ma, that the soil on till 3 developed from about 2 to 0.9 Ma, that till 4 and its paleosol formed between 0.9 and 0.5 Ma, and finally that till 5 and its paleosol developed between 0.5 Ma and about 65 ka. Karlstrom (1987, p. 2237) reported that eight samples from a silt band between paleosols 3 and 4 at the north section were reversely magnetized, which is consistent with his age assignments, since this bed should fall in the Matuyama reversed polarity zone.

The first objective of the present study was to sample all units for paleomagnetic work, in so far as was possible, and to determine their suitability as recorders of the paleofield. The second objective was to use the paleofield records to test Karlstrom's proposed chronology. If this chronology is correct, (*i*) unit 5 should belong to the Brunhes normal polarity zone and be normally magnetized, (*ii*) unit 4 should extend both below and above the Brunhes–Matuyama boundary (0.72 Ma radiometric age, 0.78 Ma astronomical age, see fig. 12) and should possess both normal and reversed magnetizations, (*iii*) unit 3 should fall in the reversed Matuyama zone (0.72 (0.78) to 2.47 (2.60) Ma) and hence be reversely magnetized, and (*iv*) units 1 and 2 should contain normal magnetizations characteristic of the Gauss normal polarity chron (older than 2.47 (2.60) Ma).

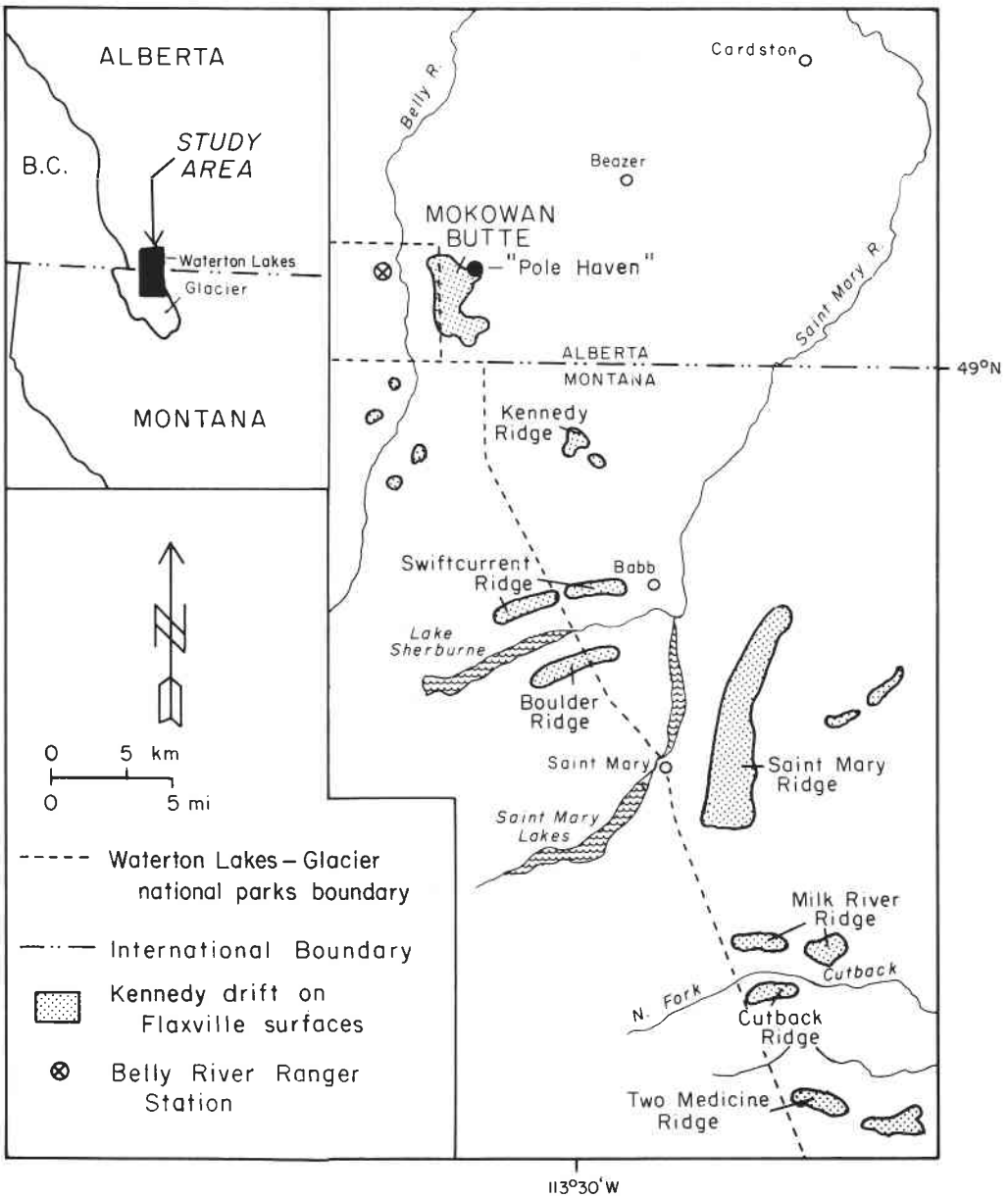


FIG. 1. Location of Mokowan Butte and other Flaxville surface remnants mantled with pre-Wisconsin drift, southwestern Alberta and northwestern Montana (modified from Karlstrom 1987).

The above discussion is somewhat simplistic because no account is taken of the possible occurrence of records of the Olduvai and Jaramillo subzones at Mokowan Butte. This simplification is justified because soil formation was probably episodic, and because it extended in any particular paleosol horizon over a considerable length of time (of the order of 100 ka) according to Karlstrom. Hence the paleosols will provide an integrated record of the paleofield that will be biased toward the recording of zones rather than subzones. We recognize that the chronology of subzones may have to be considered later, but in this, the first systematic study of paleomagnetism at Mokowan Butte, the simplified picture suffices.

Regional geology

The erosion surface on which the tills rest is a remnant of the dissected (Miocene–Pliocene) Flaxville pediment, which forms interfluves in the foothills east of the Lewis Range along the eastern margin of Waterton Lakes and Glacier national parks (Alden and Stebinger 1913; Alden 1924, 1932) (Fig. 1). This pediment is cut in Upper Cretaceous sandstone and shale and capped by 30 to over 70 m of pre-Wisconsin till, termed "Kennedy drift" (Alden and Stebinger 1913; Horberg 1956). Mokowan Butte, located about 6–10 km east of the Lewis Range, stands at an elevation of about 1800 m, or about 350 m above the Belly River to the west and 580 m above the Belly and Waterton rivers to the north. Alden (1924) postulated that the incision of this surface is largely the result of epeirogenic uplift.

At Mokowan Butte tills of the Kennedy drift consist of clasts of Precambrian Purcell Group sedimentary rocks, which outcrop in the mountains to the west, and a sandy loam matrix, which is strong brown in color. The clasts are 65–78% greenish and reddish argillite, 20–32% limestone and dolomite, 7–8% quartzite, 1–4% sandstone, and 1–4% diorite and basalt. Although Willis (1902) considered what he termed

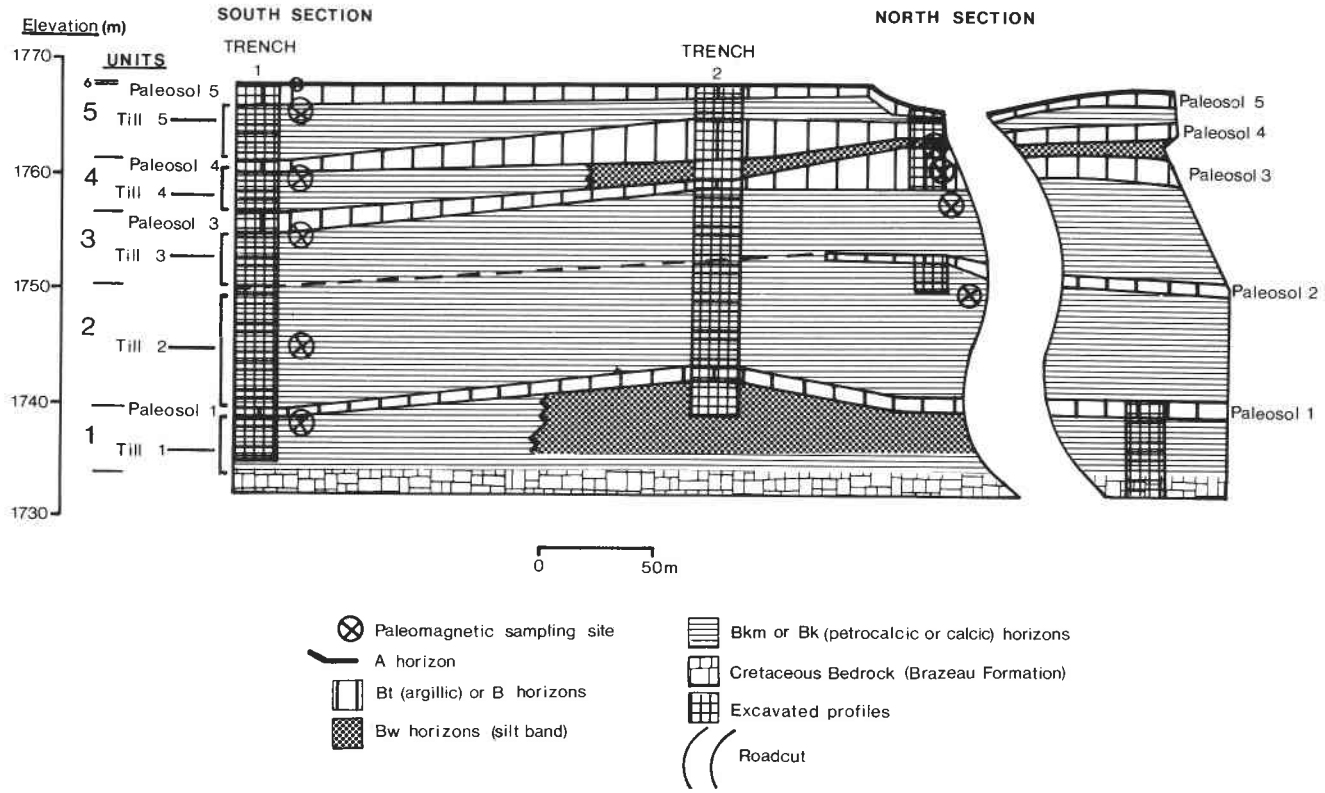


FIG. 2. Composite stratigraphic section of Kennedy drift at Mokowan Butte.

the "Kennedy gravels" to be fluvial in origin, subsequent workers (Alden and Stebinger 1913; Alden 1932; Horberg 1956; Richmond 1957, 1986; Karlstrom 1981, 1987) determined that most of the sediments within 10–12 km of the mountain front are of glacial origin. Striations are commonly preserved on greenish argillite clasts, and electron microscope analysis indicates many quartz grains have angular, possibly glacial textures (D. Krinsley, personal communication, 1986; Karlstrom 1987).

Prior to Karlstrom's work, age estimates for the Kennedy drift ranged from Nebraskan (Alden 1924) to Kansan (Horberg 1956), Illinoian (Stalker and Harrison 1977), and Early Wisconsin (Wagner 1966). It had also been suggested that the Kennedy drift may be at least partly time equivalent with the type Flaxville gravels in northeastern Montana, which include tephras that have been fission-track dated at about 7 Ma (Colton *et al.* 1983), and with the "Saskatchewan sands and gravels," which include late Tertiary to Middle Pleistocene fossils (Stalker 1983).

Richmond (1957) identified three strongly developed paleosols between drift sheets on Saint Mary and Swiftcurrent ridges (Fig. 1) and thereby subdivided the Kenney drift into three pre-Wisconsin tills, which he correlated with the Nebraskan, Kansan, and Illinoian stages of the United States midcontinent region. He later renamed these the upper, middle, and lower members of the Kennedy Formation. He also reported that the upper member yielded a uranium-trend age of 440 ± 120 ka (Richmond 1986).

Based on excavation and examination of numerous trenches across natural landslide scarps on nine Flaxville interfluves, Karlstrom (1981, 1987, 1988*a*) identified several diamicts (mostly tills) and paleosols. Of the interfluves examined, Mokowan Butte has the most complete sequence of superposed tills and the best preserved paleosols. Here, five strongly developed paleosols occur on and between five superposed till units, thus permitting subdivision of the Kennedy drift into five units (Fig. 2). Soil–till stratigraphic relations recognized in roadcut exposures examined by Karlstrom (Fig. 3) were replicated in two 30 m deep trenches located 274 m west-southwest and about 100 m southwest of the roadcut. The trenches are on a landslide scarp along the eastern edge of the butte (Fig. 2).

The buried paleosols are *in situ* and laterally continuous and represent several paleocatenas. In addition, they are morphologically, mineralogically, and chemically similar to other very strongly developed paleosols that occur throughout the region, both in Kennedy drift near the mountains and on outwash–alluvium that caps pediments east of Waterton Lakes and Glacier national parks (Karlstrom 1988*a*).

The paleosols include leached, reddish clay-rich Bt horizons (thickness 0.3–4.9 m) overlying leached B horizons and (or) calcic or petrocalcic (calcrete) BK horizons (thickness 1.5–15 m). Morphologically, they resemble subtropical-type soils (Paleudalfs, Paleustalfs, and Palexerults) found today in the southeastern and south-central United States. Hence, they presumbably formed under climates that were warmer and more moist than the present. Clasts are commonly highly weathered, but some striations are preserved on resistant greenish argillite clasts within the upper four paleosol–till units.

Based on geomorphic relations and degree of soil development, Stalker and Harrison (1977), A. MacS. Stalker (personal communication, 1984), and Karlstrom (1987, 1988*b*) concluded that the uppermost till on Mokowan Butte is early Illinoian (ca. 250–400 ka) or older in age. As already noted, the geomorphologically and stratigraphically equivalent upper till within the Kennedy Formation on Saint Mary Ridge yields a uranium-trend date of 440 ± 120 ka (Richmond 1986). Finally, their great

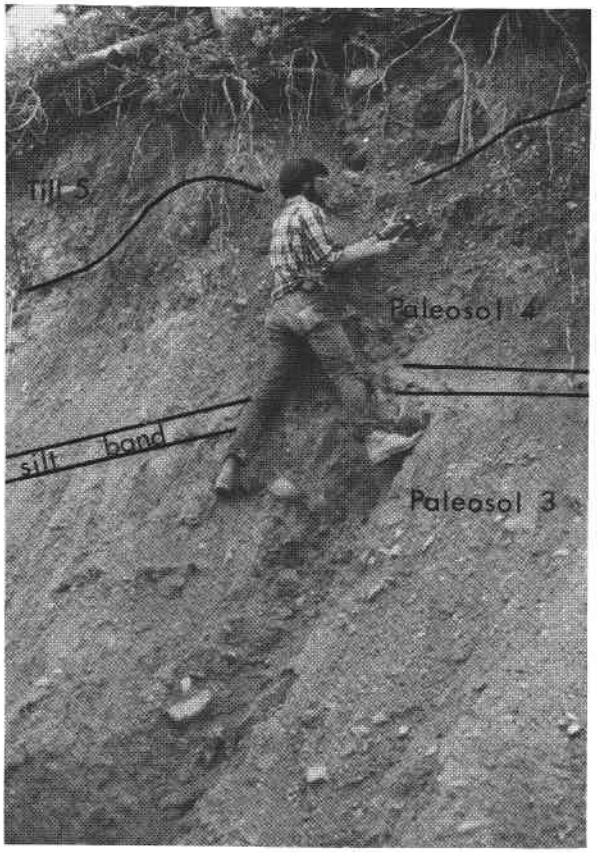


FIG. 3. Roadcut section exposing paleosol 3, silt band, paleosol 4, and till 5 on the northern edge of Mokowan Butte.

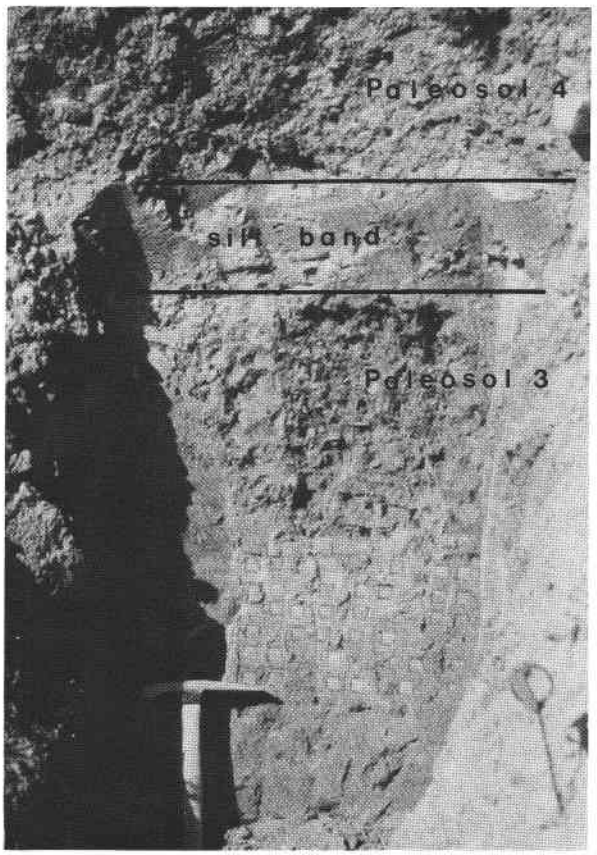


FIG. 4. Plastic sampling cubes in till 3. Paleosol 3 and silt band lie above.

thickness and degree of development suggest that each of the five Mokowan Butte paleosols spans a long period (approx. 10^5 years). This led Karlstrom (1981, 1987) to suggest that the oldest soils and associated till units could date back to the Pliocene.

Paleomagnetism

Six units from two sections, labelled north (roadcut) and south (trench), respectively, and situated along the northeastern edge of Mokowan Butte (Figs. 1–3), were sampled. In total, 125 oriented samples were collected in 2 cm × 2 cm plastic cubes (Fig. 4). The locations of the sampled units in relation to the lithostratigraphy are indicated in Fig. 2. Remanence measurements were made using automated Schonstedt spinner magnetometers. Stepwise alternating field (AF) demagnetization of pilot specimens was carried out using a Schonstedt GSD-5 for fields up to 100 mT and a Sapphire instrument SI4 for fields up to 180 mT. Demagnetization was carried out by treatment along three axes successively. All specimens were treated at several demagnetization levels. Commonly this was done in five steps, but never fewer than three and sometimes as many as twelve. The internal precision, k_s of each specimen measurement was calculated. k_s is the Fisher precision of the four estimates of direction made during measurement of each specimen. Values of k_s greater than 100 indicate that the magnetization within the specimen is approximately uniform. Values less than 5 indicate that the specimen is either very weakly magnetized and approaching the limit of detection for the instrument used, or the magnetization is very inhomogeneous, approaching randomness; such data provide no record of the paleofield. Principal magnetizations were determined either from end points or by line fitting.

Previous work by Mahaney *et al.* (1989) using scanning electron microscopy (SEM) and energy dispersive analysis has shown the presence of clay-size ferromagnetic material in these sediments. In general, the Mokowan Butte sediments are weakly magnetized, with intensities ranging from 0.1 to 0.001 A/m. The weak intensities no doubt result from the fact that the sediments covering Mokowan Butte are derived mainly from Precambrian Purcell sedimentary rocks of the Lewis Range to the west.

Unit 1(S)

Seven samples were taken from unit 1 and were collected from a fine-grained alluvial deposit contained within till 1. Their initial directions of magnetization (natural remanent magnetization or NRM) are poorly grouped; several are aligned approximately along the present Earth's magnetic field (PEF). Internal precisions (k_s) are low. Intensity declines rapidly during AF demagnetization, and no coherent high coercive force magnetizations could be isolated. The magnetizations removed after treatment in 10 mT commonly constitute as much as 80% of the original, and their directions remain grouped not far from the PEF. We concluded that the principal magnetization is a recent Brunhes-age overprint and that the original polarity of unit 1 is not determined.

Unit 2(S)

Five samples were taken from unit 2 and were collected from both till and soil. The initial directions are poorly grouped, and

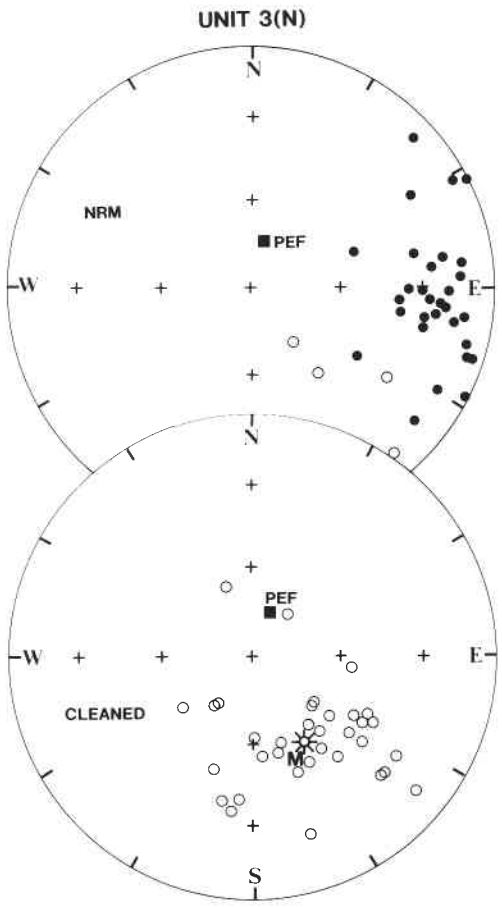


FIG. 5. Directions of magnetization, unit 3(N). NRM (top) and cleaned directions obtained after AF demagnetization in 150 mT (bottom). M is the mean direction. ● and ○, positive and negative inclinations, respectively; PEF, present Earth's field at the site. Perimeter is the present horizontal.

k_s values are about 20. Within each specimen, directions remain coherent up to 80 mT, but their grouping as a whole remains poor. After cleaning at 20 mT a mean direction of 345°, 66° (n = 5, k = 13, α_{95} = 22°) is obtained. From this evidence and the fact that only five samples were studied, it is not possible to say whether the predominantly normal magnetization is a record of the original field or an overprint.

Unit 3

This unit was sampled in detail at both the north and south sections. Samples from the north section were from both soil and till. Their NRM directions are generally easterly, with wide scatter (unit 3 (N), Fig. 5). The initial k_s values range from 15 to 100. Samples respond very well to AF demagnetization, generally showing excellent end points in the range 50–175 mT (Fig. 6). After removal of soft magnetization, directed approximately along the present field, orthogonal plots show linear decay towards the origin. All specimens were cleaned in AF of 150 mT, and the directions all become reversed. Their mean direction (M of Fig. 5) is listed in Table 1.

Samples from the southern section were also taken from both soil and till. Their NRM directions are more scattered than those at the north section (cf. Figs. 5 and 7). Initial k_s values range from 5 to 100 and are generally lower than in the north section. End points commonly occur during demagnetization, but they are less well defined than at the north section, and directions of

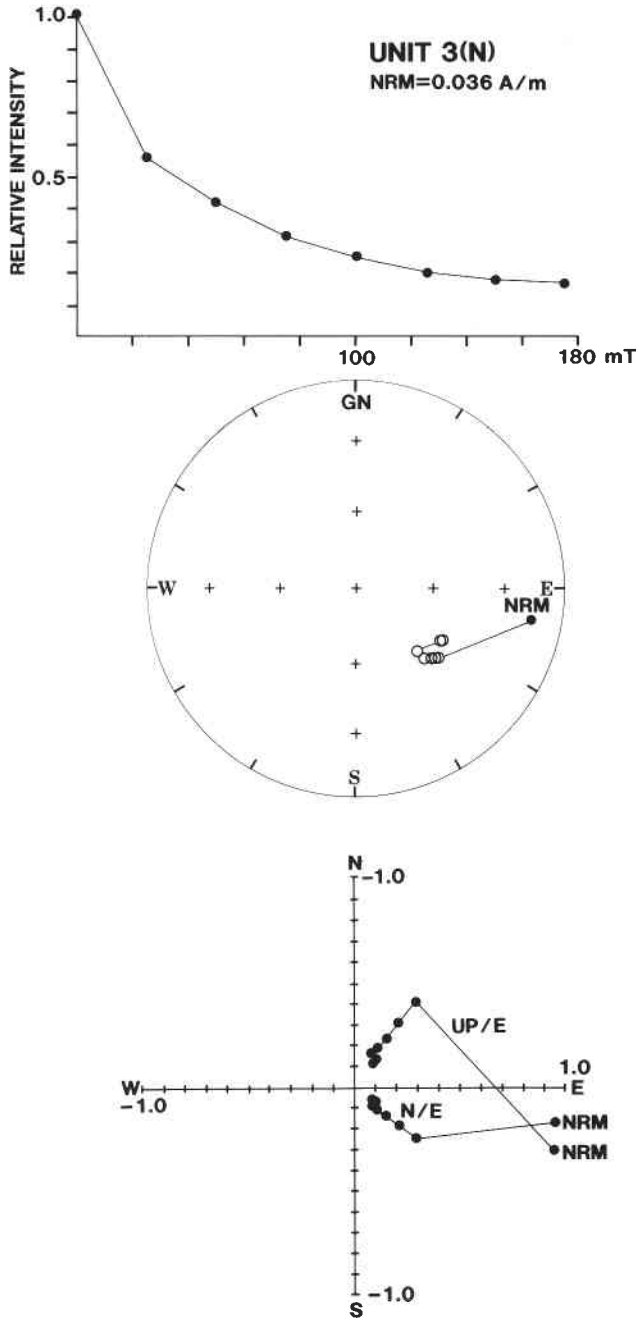


FIG. 6. Reversed magnetization of specimen from unit 3(N). Top shows decay of intensity during AF demagnetization; middle shows direction changes; bottom shows orthogonal plots.

the high coercive force magnetizations were often obtained by line fitting. The cleaned directions plotted in Fig. 7 have been obtained either from end-point determinations and cleaning at 150 mT or by line fitting in the range 15–175 mT. Most have negative inclination, but three have steep positive inclination. The mean of the former (24 values) has a negative inclination that is about 10° less than at the north section (Table 1), as if the Brunhes-age overprint has been less completely removed by the AF treatment. This perhaps is related to the fact that unit 3 in the north section is not only finer grained but is exposed in a 10 m deep roadcut and is likely, therefore, to be fresher than the same unit in the south section which was cleaned by hand to a lesser depth of about 1 m. Hematite (which is very resistant to demag-

TABLE 1. Summary of remanence directions

Unit	Polarity	n	D (°)	I (°)	k	α_{95} (°)	λ (°)	ϕ (°)
6(S)	Normal	4	348	69	55	13	82	181
5(S)	Normal	21	356	74	19	7	79	235
5 and 6	Normal	25	354	73	22	6	80	228
3(S)	Reversed	24	151	−46	6	13	59	123
3(N)	Reversed	34	149	−57	10	8	65	143
3(N and S)	Reversed	58	150	−53	8	7	63	133

NOTES: The units are shown in Fig. 22. n = number of samples; D and I = declination and inclination of the mean remanence direction; k = precision parameter; α_{95} = circle of confidence ($P = 0.05$); λ and ϕ = latitude (north) and longitude (east) of the corresponding pole.

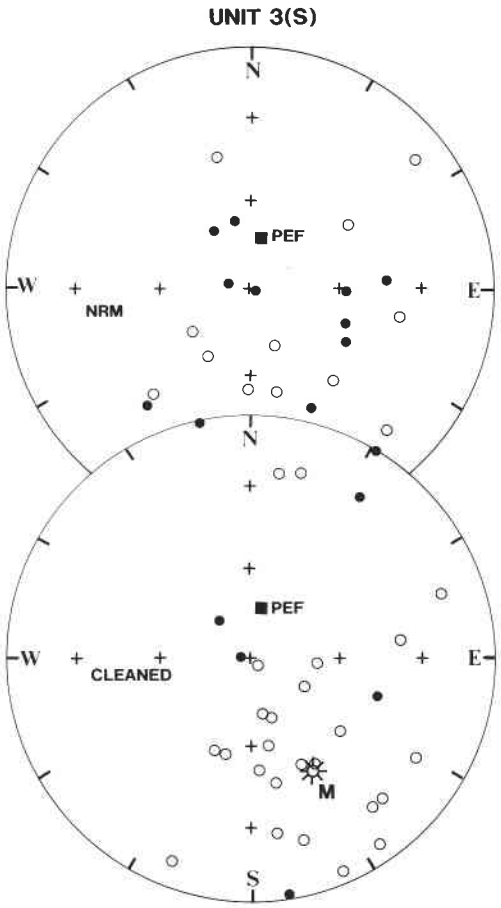


FIG. 7. Direction of magnetization, unit 3(S). Symbols as in Fig. 5. Twenty-two specimens gave end points and were cleaned in the range 125–175 mT; two specimen directions were obtained by line fitting of orthogonal plots in the range 50–175 mT.

netization), deposited by recent surface weathering, is likely to be more abundant at the south site. Such minor variations aside, however, it is clear that unit 3 is reversely magnetized.

Unit 4(S)

Samples from unit 4 were collected from both soil and till. Their NRM directions are scattered. Values of k_s range from 2 to 50. AF demagnetization of individual specimens yields coherent high coercive force magnetizations, most of which have negative inclination. Cleaning in AF of 30–175 mT gave the data of Fig. 8. Four specimens have positive inclination; these were

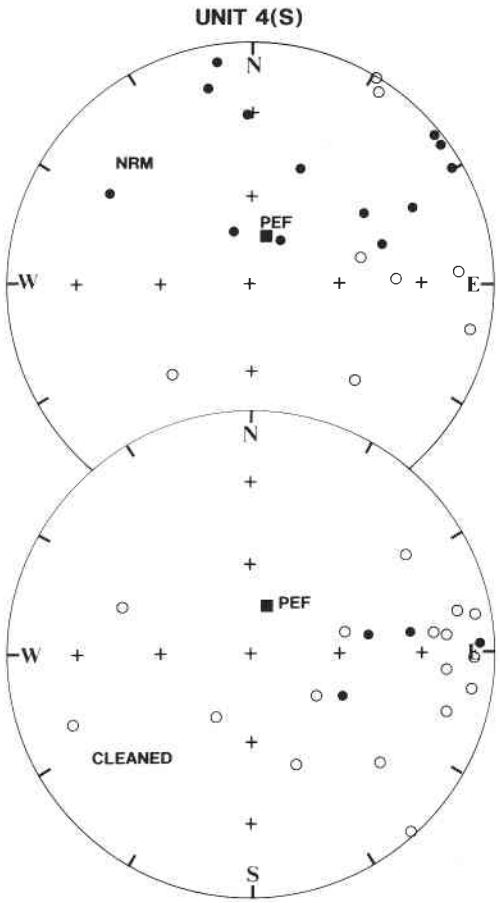


FIG. 8. Directions of magnetization, unit 4(S). Symbols as in Fig. 5. Twelve specimens gave end points and were cleaned in the range 50–175 mT; seven specimen directions were obtained by line fitting in the range 30–100 mT.

obtained at a range of levels throughout the units, thus no polarity significance can be attached to them. Clearly the cleaning process is incomplete at the AF field strengths available, and no definite polarity can be assigned to unit 4 as a whole, but the balance of the evidence would indicate that it is reversed.

Unit 5(S)

Samples from unit 5 were collected from both soil and till. This unit has NRM directions that, for the most part, are in the northwest quadrant with positive inclination (Fig. 9). Initial k_s values range from 5 to 100. Behaviour during AF demagnetization is shown in Fig. 10. End points occur, and the orthogonal plots show excellent linear decay to the origin. The principal magnetization is a high coercive force component of normal polarity with a soft, westerly directed magnetization, which is removed in fields of 5 mT. We suggest the latter is a viscous remanent magnetization (VRM) acquired during storage in the laboratory. Cleaned directions, obtained by line fitting in the range 10–100 mT or cleaning at 50 mT, are grouped around the geocentric axial dipole field (GAD), and their mean is listed in Table 1. Unit 5 has a clearly defined normal polarity, and there is no hint of reversed magnetizations.

Unit 6(S)

Four samples were collected from the E horizon of the modern soil, at a depth of 18–25 cm below the surface. Their k_s ranges

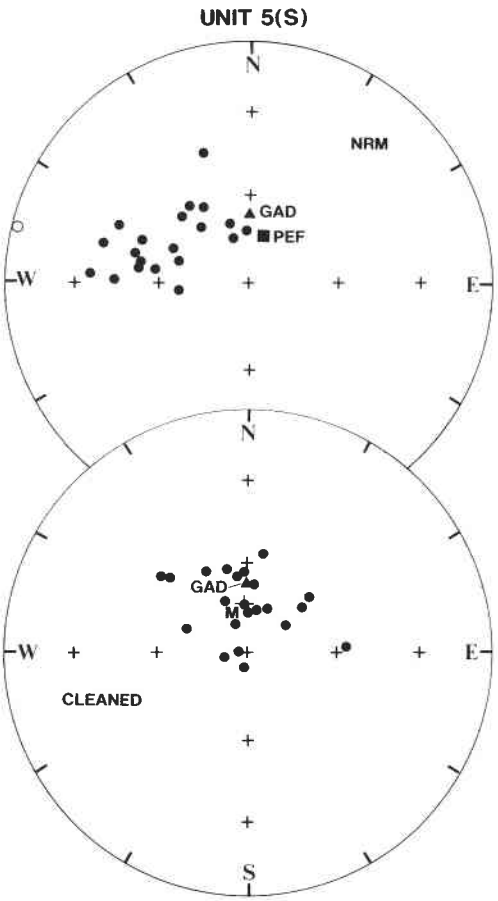


FIG. 9. Directions of magnetization, unit 5(S). Symbols as in Fig. 5. GAD, direction of the geocentric axial dipole field. Three specimens gave end points, and directions obtained at 100 mT are given; 18 specimen directions were obtained by line fitting in the range 50–150 mT.

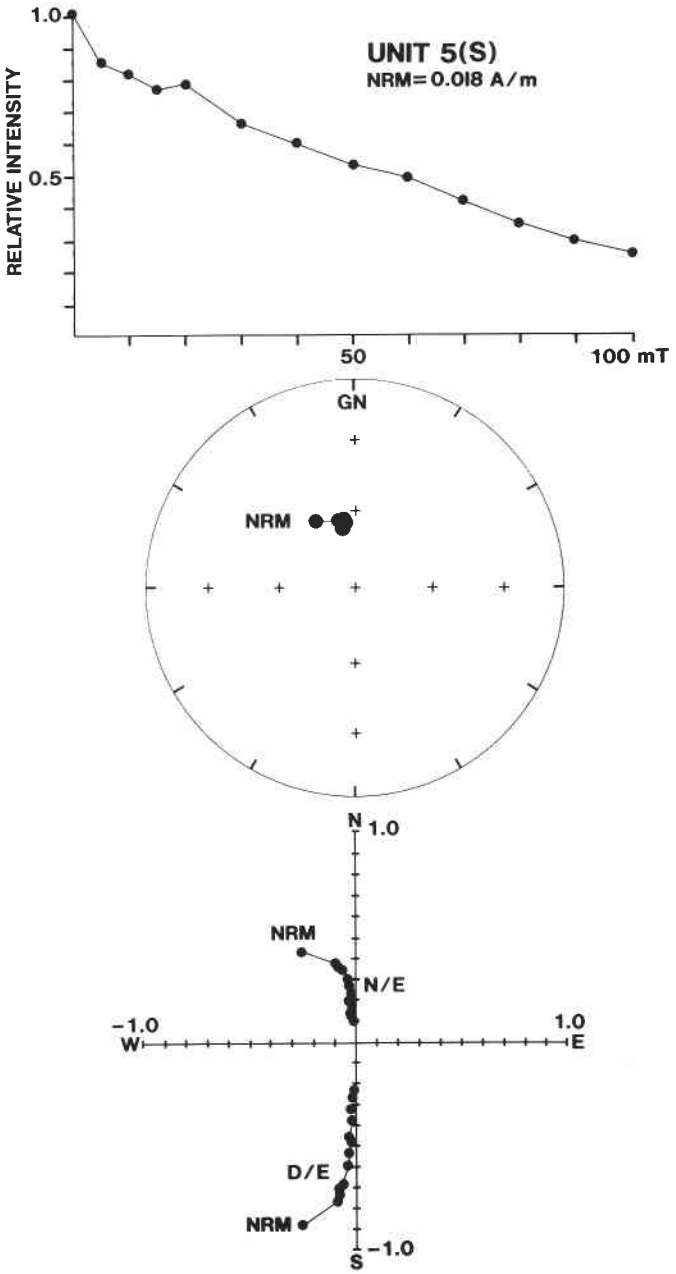


FIG. 10. Normal polarity magnetization of specimen from unit 5(S). See Fig. 6 caption.

from 10 to 60. All have well-defined principal components of normal polarity. The directions, obtained after line fitting in the range 25–100 mT, are plotted in Fig. 11, and their mean is listed in Table 1.

Discussion

Unit 3 is reversely magnetized, and units 5 and 6 are normally magnetized. Units 1, 2, and 4 are poor recorders of the field, and their polarities cannot, on the basis of this study, be defined. However, there are indications that parts of unit 4 could have reversed polarity. The mean direction of the normally magnetized tills (Table 1, units 5 and 6) are not significantly different from the GAD, and the paleopole, although nearsided, is not significantly different from the present geographical pole. The inclinations in unit 3 are significantly less than expected, and this is presumably caused by incomplete removal of Brunhes-age overprints. It is suggested that these are of recent origin and result from the presence of very high coercivity, fine-grained hematite, which is more extensive in the shallower south section than in the deeper north section (see above). However, this may not be entirely correct, as some of the overprint could be of older (but still Brunhes-age) origin because the deeper section also has a somewhat flattened inclination.

Both tills and soils in each unit are reddish to brown in color and rich in iron oxides. The passage from till to soil is gradational. Presumably iron oxides were formed initially as oxyhydroxides and then dehydrated to hematite, at which time the present magnetization was acquired. The magnetization has a very high coercive force (often exceeding 180 mT, Fig. 6) and appears therefore to be carried by hematite. It seems probable, therefore, that the magnetization of tills was, in part at least, related to and synchronous with the process of soil formation and deep weathering. However, the presence of reversed magnetization beneath normal magnetization indicates that in units 3–5 this process occurred unit by unit. If this had not been so, then the units beneath the level of reversal would have been extensively or completely overprinted by older Bruhnes-age (i.e., normal) magnetizations. As noted above, such magnetizations may be present in unit 3, but they are small and probably related to current weathering. The declination of unit 3 is to the south-southeast rather than to the south as expected. This could have a

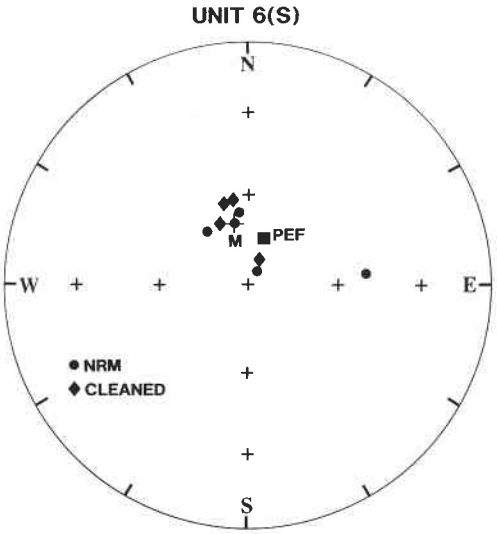


FIG. 11. Directions of magnetization, unit 6(S). Symbols as in Fig. 5. All specimen directions were obtained by line fitting in the range 25–100 mT.

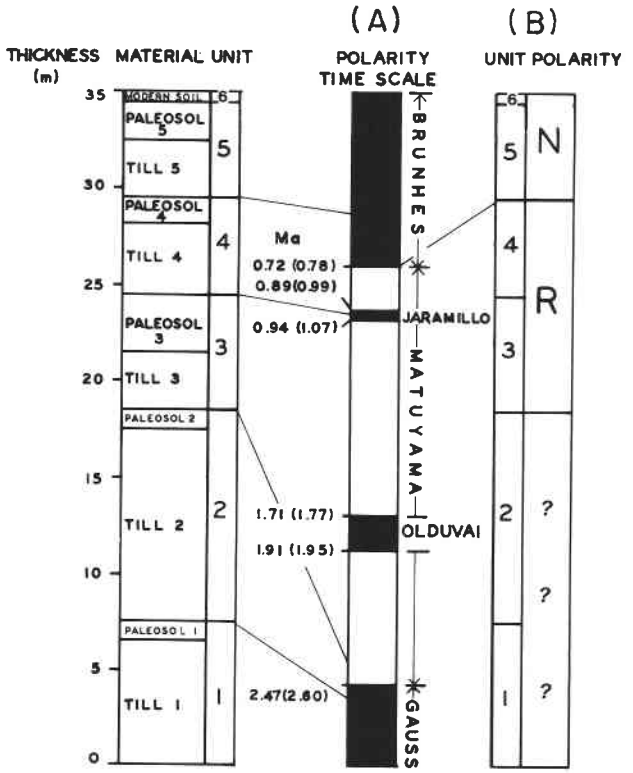


FIG. 12. Proposed chronology for Mokowan Butte sediments. (A) Chronology proposed by Karlstrom (1988) based on the Harden Index is linked to the polarity time scale (McDougall 1979). (B) Observed polarity sequence is correlated with the polarity time scale. The ages of reversal boundaries are based on radiometric determinations (McDougall 1979), with the astronomical ages in parentheses (Shackelton *et al.* 1990). The sequences sporadically record the paleofield, so the correlations proposed imply only that the studied units belong to a particular polarity zone but do not, of course, imply that they span that zone in its entirety.

number of causes, such as incomplete averaging of the paleosecular variation, incomplete removal of systematically directed Brunhes-age overprints of the type just described, or minor postdepositional tilting to the east resulting from the continuing uplift of the Rocky Mountains.

It is possible that the tills on Mokowan Butte represent some of the oldest glacial deposits in Canada. The paleomagnetic record from this site and those from Wellsch Valley, Saskatchewan (Barendregt *et al.* 1991), Banks Island, Northwest Territories (Barendregt and Vincent 1990), Fort Selkirk, Yukon (Jackson *et al.* 1990), and Merritt, British Columbia (Fulton *et al.* 1992) bring to five the number of Quaternary sequences in Canada that have reversely magnetized sediments deposited during the Matuyama chron.

Karlstrom's (1981, 1988*b*) age estimates for the Mokowan Butte sediments are shown in Fig. 12 and are compared with the polarity time scale. In general, his estimates for the younger part of the sequence are confirmed. Unit 5 is normally magnetized and entirely within the Brunhes zone, as he suggested. We have found no evidence for normal polarity (no record of the Jaramillo and Olduvai subzones) in units 3 and 4. This could mean that units 3 and 4 formed during one or other of the longer intervals of reversed polarity within the Matuyama. Alternatively, normal polarity may have been recorded initially, but during the formation of the paleosols iron minerals may have been intermittently redeposited, resulting in overprinting and loss of the original record. Sampling in units 3 and 4 was continuous, so it is unlikely that subzones, if present, were not sampled.

Data from units 1 and 2 are few and generally unsatisfactory and provide no firm polarity information. The sediments are friable and constitute poor material for magnetic studies. Further, more detailed work will be required to determine the polarities of these units reliably. Consequently, Karlstrom's assertion that glaciations in this area extend back into the Pliocene, i.e., into the Gauss normal polarity zone, cannot be tested with present data.

Acknowledgments

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ALDEN, W. C. 1924. Physiographic development of the northern Great Plains. Bulletin of the Geological Society of America, **35**: 385–424.

——— 1932. Physiography and glacial geology of eastern Montana and adjacent areas. U.S. Geological Survey, Professional Paper 174.

ALDEN, W. C., and STEBINGER, E. 1913. Pre-Wisconsin glacial drift in the region of Glacier National Park, Montana. Bulletin of the Geological Society of America, **24**: 529–572.

BARENDREGT, R. W., and VINCENT, J.-S. 1990. Late Cenozoic paleomagnetic record of Duck Hawk Bluffs, Banks Island, Canadian Arctic Archipelago. Canadian Journal of Earth Sciences, **27**: 124–130.

BARENDREGT, R. W., THOMAS, F. F., IRVING, E., BAKER, J., STALKER, A. MACS., and CHURCHER, C. S. 1991. Stratigraphy and paleomagnetism of the Jaw Face section, Wellsch Valley site, Sakatchewan. Canadian Journal of Earth Sciences, **28**: 1353–1364.

BOELSTORFF, J. 1978. North American Pleistocene stages reconsidered in light of probable Pliocene–Pleistocene continental glaciation. Science (Washington, D.C.), **202**: 305–307.

COLTON, R. B., NAESER, N. D., and WILCOX, R. E. 1983. Seven million-year-old volcanic ash on Missouri–Yellowstone River drainage divide near Circle, Montana. Geological Society of America, Abstracts with Programs, **15**: 414.

FULTON, R. J., IRVING, E., and WHEADON, P. M. 1992. Stratigraphy and paleomagnetism of Brunhes and Matuyama (>790 ka) Quaternary deposits at Merritt, British Columbia, Canadian Journal of Earth Sciences, **29**, in press.

HORBERG, L. 1956. A deep profile of weathering on pre-Wisconsin drift in Glacier National Park, Montana. Journal of Geology, **64**: 201–218.

JACKSON, L. E., BARENDREGT, R. W., IRVING, E., and WARD, B. 1990. Magnetostratigraphy of Early to Middle Pleistocene basalts and sediments, Fort Selkirk area, Yukon Territory. *In* Current research, part E. Geological Survey of Canada, Paper 90-1E, pp. 277–286.

KARLSTROM, E. T. 1981. Late Cenozoic soils of the Glacier and Waterton Parks area, northwestern Montana and southwestern Alberta, and paleoclimatic implications. Ph.D. dissertation, University of Calgary, Calgary, Alta.

——— 1987. Stratigraphy and genesis of five superposed paleosols in pre-Wisconsinan drift on Mokowan Butte, southwestern Alberta. Canadian Journal of Earth Sciences, **24**: 2235–2253.

——— 1988*a*. Multiple paleosols in pre-Wisconsinan drift, northwestern Montana and southwestern Alberta. Catena, **15**: 147–178.

——— 1988*b*. Soil development in late Quaternary glacial deposits, Waterton Park area, southwestern Alberta. Géographie physique et Quaternaire, **42**: 147–162.

——— 1991. Paleoclimatic significance of late Cenozoic paleosols east of Waterton–Glacier Parks, Alberta and Montana. Palaeogeography, Palaeoclimatology, Palaeoecology, **85**: 71–100.

MAHANEY, W. C., VORTISCH, W., BARENDREGT, R. W., and FECHER, K. 1989. Differentiation of lahars and tills by scanning electron microscopy using examples from southwestern Alberta and Mount Kenya, East Africa. Palaeoecology of Africa, **20**: 69–77.

MCDOUGALL, I. 1979. The present status of the geomagnetic polarity time scale. *In* The earth: its origin, structure and evolution. *Edited by* M. W. McElhinny. Academic Press, London, pp. 543–566.

RICHMOND, G. M. 1975. Three pre-Wisconsin glacial stages in the Rocky Mountain region. Geological Society of America Bulletin, **68**: 239–262.

——— 1986. Stratigraphy and correlation of glacial deposits of the Rocky Mountains, the Colorado Plateau and the Ranges of the Great Basin. *In* Quaternary glaciations in the Northern Hemisphere. *Edited by* V. Sibravia, D. Q. Bowen, and G. M. Richmond. Quaternary Science Reviews, **5**: 99–128.

SHACKELTON, N. J., BERGER, A., and PELTIER, W. R. 1990. An alternative astronomical calibration of the lower Pleistocene timescale based on ODP site 677. Transactions of the Royal Society of Edinburgh, Earth Sciences, **81**: 251–261.

STALKER, A. MACS. 1983. Quaternary stratigraphy in southern Alberta, Report III: the Cameron Ranch section. Geological Survey of Canada, Paper 83-10.

STALKER, A. MACS., and HARRISON, J. E. 1977. Quaternary glaciation of the Waterton – Castle River region of Alberta. Bulletin of Canadian Petroleum Geology, **25**: 882–906.

WAGNER, W. P. 1966. Correlation of Rocky Mountain and Laurentide glacial chronologies in southwestern Alberta, Canada. Ph.D. dissertation, University of Michigan, Ann Arbor.

WILLIS, B. 1902. Stratigraphy and structure of the Lewis and Livingston ranges, Montana. Bulletin of the Geological Society of America, **13**: 310.